POWER BI
FOR DATA ANALYSTS

Complete Reference Guide — 10 Sections · 200+ Techniques

Power Query · Data Modeling · DAX · Dashboards · Service · Advanced Features · Shortcuts

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Your go-to reference for every Power BI project — from raw data to published dashboard.

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How to Use This Guide

Work through each section in order for the first time — they build on each other. After that, jump directly to any section as a quick reference during your projects. Every section includes step-by-step instructions, real examples, and pro tips.

01 Introduction to Power BI

Power BI is Microsoft's business intelligence platform that transforms raw data into interactive dashboards and reports. It handles millions of rows, connects to over 100 data sources, and shares insights with your entire team — all from a free desktop application.

Power BI Desktop vs Power BI Service

Power BI Desktop	Free Windows application — where you build, design and test all reports. Connect to data, write DAX, build visuals. Cannot share directly.
Power BI Service	Cloud platform at app.powerbi.com — where you publish, share and collaborate. Schedule automatic refresh, set up security, embed reports.
Power BI Mobile	iOS and Android app — view and interact with published reports on any device. Supports touch gestures and mobile-optimised layouts.
Power BI Gateway	Bridge between on-premises data (local files, SQL Server) and Power BI Service. Required for scheduled refresh of local data sources.
Power BI Report Server	On-premises version of Power BI Service — for organisations that cannot use cloud storage due to compliance requirements.

The 6-Step Power BI Workflow

1	**Connect Data**	Get Data → choose source. Excel, CSV, SQL Server, SharePoint, Web, Folder, Azure, Dataverse and 100+ others.
2	**Clean in Power Query**	Open Power Query Editor → remove duplicates, fix data types, rename columns, split, merge, unpivot. Every step is recorded and replayable.
3	**Build Data Model**	Create relationships between tables in Model View. Use Star Schema — fact table in centre, dimension tables around it.
4	**Write DAX Measures**	Create measures for all calculations. Total Sales, Profit Margin, YTD, Growth % etc. Prefer measures over calculated columns.
5	**Design Visuals**	Build Report pages — drag fields to visuals, choose chart types, add slicers, format professionally with consistent colours.
6	**Publish & Share**	Publish to Power BI Service → share link, set up scheduled refresh, configure Row Level Security, embed or export.

■ The Most Important Habit

Always clean data in Power Query before loading. Never try to fix data quality issues with DAX after loading — it is 10x harder and your model will be slower. Clean once in Power Query, refresh forever automatically.

02 Power Query — Data Cleaning

Power Query is the data transformation engine in Power BI. Every cleaning step you perform is recorded as a step in the Applied Steps panel. When new data arrives, one click refreshes the entire pipeline automatically — repeating every transformation instantly.

15 Essential Power Query Transformations

Remove Duplicates	Table → Remove Rows → Remove Duplicates. Removes rows where all selected column values match a previous row exactly.
Remove Blank Rows	Table → Remove Rows → Remove Blank Rows. Removes any row where all cells are null or empty.
Change Data Type	Click the type icon in the column header. Set each column to the correct type — Text, Number, Date, True/False, Currency — BEFORE any other transformation.
Rename Column	Double-click any column header to rename. Use clear names with no spaces — use underscores or PascalCase for DAX compatibility.
Split Column	Transform → Split Column → by Delimiter or by Character Transition. Splits one column into multiple (e.g. "First Last" into "First" and "Last").
Replace Values	Transform → Replace Values. Find and replace text across an entire column — standardises inconsistent entries (e.g. "UK" and "United Kingdom").
Filter Rows	Click dropdown arrow on column header → apply Number, Text or Date filter conditions. Filtered rows are removed from the loaded table.
Add Custom Column	Add Column → Custom Column → write M formula. Example: = [Sales] * [Margin Rate] — creates a new calculated column using M language.
Merge Queries	Home → Merge Queries → select second table and matching key column → choose join type (Left, Inner, Full, Anti). Power BI's version of VLOOKUP.
Append Queries	Home → Append Queries → stack two or more tables vertically. Use to combine monthly files or sheets into one dataset automatically.
Unpivot Columns	Select columns to unpivot → Transform → Unpivot Columns. Converts wide data (one column per month) into tall data (Month column + Value column).
Group By	Transform → Group By → choose grouping column and aggregation. Like a PivotTable embedded in the query — aggregates before loading.
Fill Down / Up	Transform → Fill → Down or Up. Fills blank cells with the value from the previous (or next) non-blank cell. Fixes merged cell exports.

Trim & Clean	Transform → Format → Trim removes leading/trailing spaces. Clean removes non-printable characters. Apply to all text columns from external data.
Promote Headers	Use First Row as Headers — promotes the first data row to become column headers. Essential when importing files where headers are in row 1.

Key M Language Patterns

Function	Syntax	What It Does
Table.RemoveDuplicates	= Table.RemoveDuplicates(Source)	Remove duplicate rows
Table.SelectRows	= Table.SelectRows(Source, each [Sales] > 0)	Filter rows by condition
Table.AddColumn	= Table.AddColumn(Source, "Margin", each [P]/[S])	Add calculated column
Text.Trim	= Table.TransformColumns(Source,{{"Col",Text.Trim}})	Trim spaces from text column
Date.Year	= Table.AddColumn(Source, "Year", each Date.Year([D]))	Extract year from date
Table.Combine	= Table.Combine({Table1, Table2})	Stack two tables vertically
Table.Unpivot	= Table.Unpivot(Source, {"Jan","Feb"}, "Month","Val")	Unpivot named columns

■ Record First, Edit Second

Never write M code from scratch. Perform transformations using the Power Query UI buttons and ribbon. Power BI records every click as M code in Applied Steps. Then open Advanced Editor to study and modify the generated code. This is the fastest way to learn M Language while solving real problems.

03 Data Modeling

Data modeling defines how your tables connect and how filters flow between them. A correct data model makes your DAX simpler, your reports faster and your filters work accurately every time. The gold standard is the Star Schema.

Star Schema — The Gold Standard

Fact Table	The central table containing measurable events — sales transactions, orders, page views, employee records. Has many rows. Contains numeric values and foreign keys.
Dimension Tables	Lookup tables surrounding the fact table — Date, Product, Customer, Region, Employee. Has fewer rows. Contains descriptive attributes used for filtering and slicing.
Relationships	Connections between tables on matching key columns. Always one-to-many (1:*) — one row in the dimension matches many rows in the fact table.
Filter Direction	Single direction — filters flow FROM dimension tables TO the fact table. This is the default and correct setup for 95% of reports.
Date Table	A dedicated dimension table with one row per date, covering your full data range. MUST be marked as Date Table for time intelligence DAX to work.

10 Data Modeling Rules

1	**Always use Star Schema**	One fact table in the center, dimension tables surrounding it on all sides.
2	**Always create a Date Table**	Never use dates directly from your fact table — create a separate Dim_Date table.
3	**Mark your Date Table**	Right-click the date table → Mark as Date Table → select the Date column. Required for time intelligence DAX.
4	**Use integer surrogate keys**	Relate tables on integer ID columns, not text. Integer joins are significantly faster in memory.
5	**One-to-many relationships only**	Avoid many-to-many (M:M) relationships — they cause ambiguous filters and incorrect totals.
6	**Single filter direction**	Default single-direction filters (dimension → fact) are faster and more predictable. Avoid bi-directional unless essential.
7	**Hide all key columns**	Right-click key/ID columns → Hide in Report View. Users should only see descriptive columns, not surrogate keys.
8	**Name tables and columns clearly**	Use Fact_Sales, Dim_Date, Dim_Customer. Rename columns — "CustomerID" not "CustID_FK". Good names make DAX readable.
9	**Avoid circular relationships**	A chain of relationships that loops back to the starting table — Power BI will error. Redesign your model structure.

10	**Keep the model flat**	Fewer tables with more columns is better than many thin tables. Avoid snowflaking dimension tables unless absolutely necessary.

■ How to Create a Date Table in DAX

In the Modeling tab → New Table, then enter: Date = CALENDAR(MIN(Fact_Sales[OrderDate]), MAX(Fact_Sales[OrderDate])). Then add columns: Year = YEAR([Date]), Month = MONTH([Date]), MonthName = FORMAT([Date],"MMMM"), Quarter = "Q" & QUARTER([Date]). Finally right-click the table → Mark as Date Table → select the Date column.

04 DAX — Basics

DAX (Data Analysis Expressions) is the formula language used for all calculations in Power BI. It operates on entire columns and tables rather than individual cells — making it fundamentally different from Excel formulas.

Calculated Columns vs Measures

	Calculated Column	Measure
Where stored	In the table — physically added as a new column	Not stored — calculated on demand only
When calculated	At data refresh — once per row	At query time — responds to every filter change
Memory usage	Uses RAM — increases model size	Memory efficient — no model size increase
Filter aware	No — calculated during refresh, ignores slicers	Yes — recalculates for every filter combination
Use for	Columns needed for relationships or row filtering	All KPIs, totals, percentages and visual values
Example	[Delivery Days] = [ShipDate] - [OrderDate]	[Total Sales] = SUM(Sales[Revenue])

Filter Context and Row Context

Row Context	The current row being evaluated in a calculated column or iterator function. Each row has its own context — [Sales] * [Margin] works row by row.
Filter Context	The set of filters currently applied by slicers, visual interactions and CALCULATE. Measures always evaluate inside the current filter context.
CALCULATE	The most powerful DAX function — it evaluates an expression inside a MODIFIED filter context. CALCULATE([Total Sales], Region[Region]="East") changes the filter context before evaluating.
Context Transition	When a measure is evaluated inside a row context (e.g. inside SUMX), the row context is automatically converted into an equivalent filter context.

DAX Syntax Rules

Table[Column]	Reference a column: Sales[Revenue] — always use TableName[ColumnName] format for clarity.
[MeasureName]	Reference a measure: [Total Sales] — measures are referenced with brackets only, no table name.
// comment	Single-line comment — use // to add notes inside DAX formulas for documentation.

VAR / RETURN	Variable syntax — VAR x = [Total Sales] / 100 RETURN x — stores intermediate results for readable formulas.
DIVIDE(a,b,0)	Always use DIVIDE() instead of / — the third argument is the value to return if dividing by zero (usually 0 or BLANK()).

■ The Golden Rule of DAX

Default to Measures for everything. Only create a Calculated Column when you genuinely need a column for row-level filtering or a relationship. A model with 50 measures and 3 calculated columns will always outperform a model with 20 measures and 30 calculated columns.

05 DAX — Functions & Measures

Aggregation Functions

Total Sales	`SUM(Sales[Revenue])`	Sums all values in the Revenue column across current filter context.
Total Orders	`COUNTROWS(Sales)`	Counts total rows in the Sales table — each row = one order.
Unique Customers	`DISTINCTCOUNT(Sales[CustomerID])`	Counts unique customer IDs — ignores duplicates.
Avg Order Value	`AVERAGE(Sales[Revenue])`	Average revenue per order in the current filter context.
Max Sale	`MAX(Sales[Revenue])`	Highest single revenue value in current filter.

CALCULATE & Filter Functions

East Sales	`CALCULATE([Total Sales], Region[Region]="East")`	Overrides Region filter — always shows East regardless of slicer.
Sales Excl Returns	`CALCULATE([Total Sales], FILTER(Sales, Sales[Type]<>"Return"))`	Removes return orders from the calculation.
All Products Sales	`CALCULATE([Total Sales], ALL(Product))`	Removes ALL filters on the Product table — shows grand total.
Selected Product	`CALCULATE([Total Sales], ALLEXCEPT(Product, Product[Name]))`	Keeps only the Product Name filter, removes everything else.
High Value Orders	`CALCULATE([Total Orders], Sales[Revenue]>1000)`	Counts only orders where revenue exceeds 1,000.

Ratio & Percentage Measures

Profit Margin %	`DIVIDE([Total Profit],[Total Sales],0)`	Safe division — returns 0 if Total Sales is zero or blank.
% of Total	`DIVIDE([Total Sales],CALCULATE([Total Sales],ALL(Sales)),0)`	Each category as a percentage of the grand total.
Growth %	`DIVIDE([Total Sales]-[Sales LY],[Sales LY],0)`	Year-over-year growth percentage.
Sales Rank	`RANKX(ALL(Product[Name]),[Total Sales],,DESC)`	Ranks each product by Total Sales, largest = 1.

Time Intelligence Functions

Sales LY	`CALCULATE([Total Sales],SAMEPERIODLASTYEAR(Date[Date]))`	Same period in the previous year — requires marked Date Table.
Sales YTD	`TOTALYTD([Total Sales],Date[Date])`	Cumulative total from Jan 1 to the last date in filter.
Sales MTD	`TOTALMTD([Total Sales],Date[Date])`	Cumulative total from first of month to last date in filter.
Sales QTD	`TOTALQTD([Total Sales],Date[Date])`	Cumulative total from first of quarter to last date in filter.
Rolling 3M Sales	`CALCULATE([Total Sales],DATESINPERIOD(Date[Date],LASTDATE(Date[Date]),-3,MONTH))`	Last 3 months rolling window.

■ CALCULATE is the Foundation of Advanced DAX

Almost every complex DAX measure uses CALCULATE. It does one thing — evaluates an expression in a modified filter context. Master these three patterns: (1) CALCULATE with a direct filter, (2) CALCULATE with ALL to remove filters, (3) CALCULATE with time intelligence functions. These three patterns cover 80% of all real-world DAX needs.

06 Data Visualization

The right visual makes data instantly understandable. The wrong visual creates confusion. Choose chart types based on the relationship you are showing — not based on what looks impressive.

12 Visual Types — When to Use Each

Visual	Best For	Avoid When
Bar / Column Chart	Comparing values across categories (regions, products, months)	Too many categories — bars become unreadably thin
Line Chart	Showing trends and changes over time (monthly, quarterly, yearly)	Data has no time dimension or natural sequence
Area Chart	Showing cumulative totals or volume over time	Comparing multiple series — areas overlap and hide data
Pie / Donut Chart	Showing part-to-whole with 3-4 categories maximum	More than 4 segments — slices become too small to read
Scatter Chart	Showing correlation or relationship between two numeric variables	Categorical data — scatter needs two numeric axes
Matrix	Showing data in rows × columns with subtotals and drill-down	Simple comparisons — a bar chart communicates faster
Card	Displaying a single KPI number prominently (Revenue, Count, %)	Multiple numbers — use a table or matrix instead
Gauge	Showing progress toward a specific target value	No defined target — gauge without context is meaningless
Treemap	Showing hierarchical part-to-whole with size and colour encoding	Precise comparisons — areas are hard to compare accurately
Map / Filled Map	Showing geographic distribution of data by location or region	Non-geographic categories — use bar chart instead
Waterfall Chart	Showing how values build up or break down from a starting point	Simple totals — a bar chart is clearer and faster to read
Funnel Chart	Showing stages in a sequential process — sales pipeline, onboarding	Data with no natural sequence or ranking

Formatting for Professional Reports

Insight-first titles	Never write "Sales by Region" — write "West Region Leads Revenue at 38% of Total". Titles should tell the story, not describe the chart.
Remove gridlines	Click any gridline → Delete. Clean charts communicate more clearly. Gridlines add visual noise without adding analytical value.

Consistent colours	The same category must always use the same colour across all visuals on all pages. Create a consistent colour theme at report level.
Data labels	Only label the most important values — labelling everything creates clutter. Use data labels sparingly on bar charts and KPI cards.
Remove borders	Right-click visual → Format → Visual border → Off. Borderless visuals look modern and professional. Use spacing instead of borders.
Align to grid	Select multiple visuals with Shift+click → Format → Align. Pixel-perfect alignment communicates professionalism and builds trust.
Conditional formatting	Format → Conditional Formatting → colour scale or rules. Automatically colour cells green/red based on thresholds — no manual work.

■ The Most Overlooked Feature — Conditional Formatting

Conditional formatting transforms a plain matrix into a visual heatmap automatically. Select any value column in a matrix → Format → Conditional Formatting → Background Color → choose a color scale. Highest values go green, lowest go red — instant visual pattern recognition without building a separate chart.

07 Dashboard Design

A great dashboard answers one question per page, instantly, without explanation. Before building, write one sentence: what decision does this dashboard support? Every single element must serve that purpose.

15-Point Dashboard Design Checklist

Layout

✓	**One clear purpose per page**	Write it before you build: "What is the sales performance by region this quarter?"
✓	**KPI cards at the top**	Key numbers always visible first — Total Sales, Total Orders, Margin %, Growth %.
✓	**Main charts in the center**	The largest, most important visual gets the most prominent position on the canvas.
✓	**Filters and slicers on the side**	Consistent position — right panel or left panel. Never change position between pages.
✓	**All visuals aligned to the grid**	Select all → Format → Align. Use the gridlines. Misaligned visuals signal sloppiness.

Design

✓	**Maximum 3 colours in your scheme**	Primary (action/highlight), secondary (supporting), neutral (background). No more.
✓	**Consistent font sizes throughout**	Page title 18pt, card titles 14pt, visual titles 12pt, labels 10pt. Never deviate.
✓	**White or dark background only**	No gradients, no textures, no patterned backgrounds. Plain white or plain dark.
✓	**Remove all unnecessary borders**	Visual borders, grid borders, table borders — remove them all. Use spacing instead.
✓	**Data labels on key values only**	Label the top 3 bars, the biggest slice, the KPI number. Not everything.

Content

✓	**Every visual has a clear title**	Insight-first titles — not "Sales by Region" but "West Leads with 38% of Revenue".
✓	**Axis labels are readable**	Never rotate labels 90 degrees. Use horizontal bar charts if category names are long.
✓	**Numbers formatted correctly**	Currency uses £/$ symbol. Percentages use %. Large numbers use K/M suffix (1.2M not 1200000).
✓	**All slicers connected to visuals**	Right-click slicer → Edit Interactions — ensure all visuals respond. Test every combination.
✓	**Tested on mobile layout**	View → Mobile Layout — check report is usable on phone. Stack visuals vertically for mobile.

■ Dashboard vs Report — Know the Difference

In Power BI, a Report is a multi-page file you build in Desktop (.pbix). A Dashboard is a single page in Power BI Service built by pinning visuals from multiple reports. Dashboards do not support page-level filters or slicers. For most analysts, the Report is where you spend 95% of your time — the Dashboard is for executive summary pinboards.

08 Power BI Service

Publishing Reports

1	**Save your .pbix file**	File → Save. Always save before publishing to ensure the latest version is uploaded.
2	**Click Publish**	Home tab → Publish button. Power BI Desktop will ask you to sign in if not already.
3	**Choose workspace**	Select the workspace to publish to. "My Workspace" for personal use. Shared workspace for team access.
4	**Wait for upload**	Larger files with many visuals take longer. A success message confirms when done.
5	**Open in Service**	Click the link in the success dialog to open the report directly in Power BI Service in your browser.

6 Ways to Share — Which to Use When

Share a Link	Simplest method. Click Share → enter email or copy link. Recipient needs Power BI Pro license to view. Best for daily team sharing.
Publish to Web	Creates a public URL viewable by anyone with no login. WARNING — anyone with the link can see the data. Use ONLY for public, non-sensitive data.
Export PDF/PPT	File → Export → PDF or PowerPoint. Creates a static snapshot — no interactivity. Best for executives who do not use Power BI directly.
Embed in SharePoint	Add Power BI web part to a SharePoint page. Report stays interactive inside SharePoint. Best for internal team portals and intranets.
Embed in Website	File → Embed Report → Website or Portal → copy the iframe code. Best for customer-facing portals. Requires Premium capacity or per-user Premium.
Power BI App	Bundle multiple reports and dashboards into one app. Publish app to a group. Best for delivering a complete analytics package to a team or department.

Setting Up Scheduled Refresh

1	**Open dataset settings**	In your workspace → find the dataset (not the report) → click three dots (■) → Settings.
2	**Expand Scheduled Refresh**	Toggle "Keep your data up to date" to ON. The refresh schedule options will appear below.
3	**Set frequency**	Choose Daily, Weekly or specific days. Set one or more refresh times (up to 8 per day on free, 48 on Premium).

4	**Configure Gateway**	Required if your data source is on-premises (local Excel files, SQL Server). Install Power BI Gateway on the computer hosting the data.
5	**Enter credentials**	Expand Data Source Credentials → Edit → enter username and password for each data source.
6	**Click Apply**	Save settings. Power BI Service will now refresh automatically at the scheduled times.

■ Enable Refresh Failure Notifications

In Settings → Scheduled Refresh → scroll down to "Send refresh failure notifications" → set to Report Owner. You will receive an email if any scheduled refresh fails — before your team notices the data is stale. Also check the Refresh History in dataset settings to see success/failure logs with error messages.

09 Advanced Features

Row Level Security (RLS)

RLS restricts which rows of data each user can see in the same report — one report serves everyone, each person sees only their data.

1	**Create role**	Modeling tab → Manage Roles → New → name the role (e.g. "East Region").
2	**Write DAX filter**	Select the table → write a DAX filter: [Region] = "East" or [Email] = USERPRINCIPALNAME() for dynamic RLS.
3	**Test in Desktop**	Modeling → View As → select your role → verify the report filters correctly before publishing.
4	**Publish to Service**	Publish normally. The roles are included in the .pbix file.
5	**Assign users to roles**	In Power BI Service → dataset → Security → select role → add user email addresses.

Bookmarks

What they do	Save the complete state of a report page — all filter values, slicer selections, visible/hidden visuals — as a named snapshot.
Create	View → Bookmarks → Add. Rename each bookmark clearly: "All Regions", "East Only", "Default View".
Use cases	Navigation buttons that change the view, toggle between chart types, show/hide detail panels, create guided story presentations.
With buttons	Insert → Button → choose type → in the Action section → set Type to Bookmark → select your bookmark. Now the button applies that saved state on click.
Report sections	Create separate bookmarks for each section of a long report — use a left-side navigation panel with bookmark buttons to simulate multi-page navigation.

Drillthrough

What it does	Right-clicking a data point in any visual opens a detail page filtered to that exact item — product, customer, region, etc.
Set up	Create a new report page → drag the drillthrough field (e.g. Product[Name]) to the Drillthrough well in the Visualizations panel.
How to use	Right-click any bar, data point or table row in the main report → Drill through → select the detail page name.
Back button	Power BI automatically adds a Back button to drillthrough pages — users can return to the source page with one click.

5 More Advanced Features

Decomposition Tree	AI visual — automatically breaks down a metric (Total Sales) by any dimension you drag in. Drag in Region, then Category, then Product to drill down step by step.
Smart Narrative	AI visual — automatically generates a text summary of your data. Updates dynamically when filters change. Add to any page for instant written insights.
Q&A; Visual	Natural language query box — users type "show sales by region as bar chart" and Power BI builds it instantly. Great for executive dashboards.
Key Influencers	AI visual — analyzes which columns most influence a target metric (e.g. what factors drive customer churn). Uses machine learning automatically.
Field Parameters	Let report users switch which measure or dimension a visual shows — without you creating separate pages. A single slicer changes the entire visual.

■ Dynamic RLS — One Setup for All Users

Static RLS requires you to manually assign each user to a role (East, West, North etc). Dynamic RLS requires only one role. Create a Users table with Email and Region columns. Write this DAX filter on your dimension table: [Region] = LOOKUPVALUE(Users[Region], Users[Email], USERPRINCIPALNAME()). Now Power BI automatically filters each user to their region based on their login email. No manual role assignments needed.

10 Shortcuts & Quick Reference

General

Ctrl + S	Save the current .pbix file
Ctrl + Z	Undo last action
Ctrl + Y	Redo last action
Ctrl + C / V	Copy / Paste selected visual
Ctrl + D	Duplicate selected visual
Ctrl + G	Group selected visuals
F5	Start Presentation / Full screen mode
Esc	Exit full screen or cancel current action

Canvas & Visuals

Ctrl + A	Select all visuals on the current page
Ctrl + Click	Select multiple individual visuals
Arrow keys	Nudge selected visual(s) by 1 pixel
Shift + Arrow	Nudge selected visual(s) by 10 pixels
Ctrl + Shift + G	Ungroup selected group of visuals
Ctrl + F	Find & replace text in DAX editor
Tab	Move focus to next visual on page
Delete	Delete selected visual(s)

Report Navigation

Ctrl + Page Down	Go to next report page
Ctrl + Page Up	Go to previous report page
Ctrl + Shift + F	Open Fields pane
Ctrl + Shift + P	Open Format pane
Ctrl + Shift + A	Open Analytics pane
Ctrl + Shift + V	Open Visualizations pane
Alt + Shift + F1	New measure (in Data/Model view)
Alt + Shift + F2	New calculated column (in Data view)

Power Query Editor

Ctrl + P	Open Power Query Editor from Desktop
Ctrl + Enter	Confirm a formula in the formula bar
Ctrl + Z	Undo last transformation step
Delete (step)	Click step in Applied Steps → Delete to remove
F2	Rename selected query in the Queries pane
Ctrl + T	View M code in Advanced Editor
Ctrl + C (step)	Copy an Applied Step to paste into another query
Right-click column	Context menu with all transform options for that column

DAX Editor

Ctrl + Space	Trigger DAX IntelliSense — shows function/column suggestions
Tab	Accept current IntelliSense suggestion
Shift + Enter	New line inside a DAX formula (multi-line formatting)
Ctrl + /	Comment out selected line(s) with //
Ctrl + K, Ctrl + F	Auto-format / indent the DAX formula
Ctrl + Home	Jump to start of formula
Ctrl + End	Jump to end of formula
Ctrl + A	Select all text in DAX editor

■ Performance Tips

If your report feels slow: (1) Reduce the number of visuals per page — each visual runs a separate DAX query. (2) Remove unused columns and tables from your model — every column loaded into memory costs RAM. (3) Replace calculated columns with measures where possible. (4) Use Import mode not DirectQuery unless real-time data is essential — Import is 10x faster. (5) Avoid bi-directional relationships — they generate complex queries that slow everything down.

Power BI is how you tell it.

- → Build your first real Power BI project using a dataset you care about
- → Learn Power Query first — clean data is the foundation of every great report
- → Master CALCULATE — it unlocks 80% of advanced DAX patterns
- → Follow the Star Schema rule on every single project without exception
- → Share your work publicly on LinkedIn — feedback accelerates learning faster than anything
- → Follow for weekly Power BI tips, DAX patterns and real-world project walkthroughs

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